

Guideline Title: Extracorporeal Membrane Oxygenation Therapy (ECMO)

Guideline Number: GUID-3203-C008

Issue date: 6/4/2021	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
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- I. **PURPOSE:** Extracorporeal membrane oxygenation (ECMO) therapy is an invasive strategy for treatment of severe pulmonary and/or cardiac disease utilizing a “heart-lung” bypass device to allow for gas exchange. Patients requiring ECMO are critically ill and resource intensive, requiring specialized care and equipment. Patients are cannulated by surgeons and usually managed by perfusionists who have expertise in the device. Venous-Arterial (VA) ECMO “bypasses” both the heart and lungs, and is used for severe cardiogenic shock, Venous-Venous (VV) ECMO “bypasses” the lungs only and is used for acute respiratory distress syndrome, massive pulmonary embolism, pulmonary contusions, and other issues.
- II. **DEFINITIONS:**
 - A. ECMO – Extracorporeal Membrane Oxygenation
- III. **DEPARTMENT GUIDELINE:**
 - A. **EQUIPMENT:**
 - i. Cardiohelp console (~22 lbs.)
 - ii. Two ECMO Circuits (~10 lbs.) – one for primary, one for backup. (If patient is already on Cardiohelp console, we will only bring one as a backup)
 - iii. Transport backup (~15 lbs.) – contains tubing, connectors, clamps, IV lines, prep sticks etc.
 - B. **INDICATIONS:**
 - i. Acute respiratory or cardiovascular deterioration unresponsive to therapy
 1. Rarely, this includes prolonged resuscitation during cardiac arrest
 - C. **PROCEDURE:**
 - i. Establish care defined by GUID3203-G002 – *General Patient Care*.
 - ii. ECMO Specialist (perfusionist or trained RN)
 1. Primary responsibility prior to transport: ensure all equipment and blood products on checklist are available, functional, and loaded at time of departure
 2. Primary responsibility during transport: management of ECMO circuit during all phases of transport
 - iii. Medical Flight Crew

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1. Patient assessment
 - a. Assess extremity blood flow
 - b. Assess all lines, tubes, and catheters
2. Primary responsibility during transport: non-circuit related clinical care
3. Administration of medications, fluids, and blood products per *GUID3203-G002 Management of previously initiated and continuous IV Medication*, *GUID3203-G005 Analgesia and sedation management*, *GUID3203-G007 Propofol (Diprivan)*, *GUID3203-PR025 Emergency blood product administration*, and *GUID3203-PR026 Warming of blood products and intravenous fluids*
 - a. Consider changing sedation and pain medication to less protein-bound and lipophilic drugs (i.e., Ativan and Diluadid if available).
4. Ventilator set-up, management and troubleshooting per *GUID3203-PR024 Ventilation with a mechanical ventilator*
 - a. If patient is being administered aerosolized **Reproterol (Flolan)** and the appropriate equipment is available, continue at current dosing in consultation with sending/admitting physician or Medical Control
5. If present, IABP set-up, management and troubleshooting per *GUID3203-PR023 Intra-aortic balloon pump*.
6. Assist with monitoring and blood gas analysis
7. Treat hypotension per *GUID3203-M011 – Hypotension* in consultation with perfusionist and accepting physician.

D. TROUBLE SHOOTING

- i. ECMO system trouble shooting performed by ECMO perfusionist
- ii. Ventilator, Monitor, IV Pumps and any additional equipment troubleshooting performed by Medical Flight Crew.

IV. DOCUMENTATION: EMS Charts

- A. Q5 minute vitals
 - i. Heart Rate, Blood Pressure (Arterial or NIBP), ETCO₂, SPO₂, and Respiratory rate
- B. ECMO system values charted at:
 - i. Initiation of care
 - ii. Transfer of care
 - iii. Change in patient condition
 1. Blood Flow Rate, Pump Speed, Circuit speed, Temperature
- C. ECMO patient specific assessment charting.
 - i. ECMO catheter location and appearance

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- ii. Additional monitor lines
- iii. IV access lines

V. REFERENCES:

- A. Extracorporeal Life Support Organization (ELSO) Guidelines for ECMO Transport
- B. Orlando Health ECMO Criteria Guideline GUID-1620-0900